

WASSWA WILSON

Biomedical Engineer · Full-Stack Software Developer · Health Technology Programme Lead

Kampala, Uganda · wasswawilson0001@gmail.com · +256 783 230 321
English (fluent, professional working proficiency) · Luganda (native)

PROFESSIONAL PROFILE

I am a biomedical engineer and full-stack software developer based in Kampala, with over nine years of work across hospital technology, health information systems and software engineering. I lead development of the FairBanks Community Health Intelligence Platform (FCHIP), where I handle the data capture tools, the database, the cloud data flows and the management dashboards. Alongside it I am building a cross-sector application that collects field data and generates finished reports in real time, for accounting, research, biology, agriculture, social media and other record-heavy work. Before FairBanks I managed the Gould Family Foundation biomedical programme across East and Central Africa, and ran the biomedical function at International Hospital Kampala through the COVID-19 response and through successful COHSASA accreditation. I write Java, Python, JavaScript and TypeScript, work across React, Next.js, React Native, Node.js and Express, and design AI agents and automations that are running in production today.

CAREER HIGHLIGHTS

- Lead developer of FCHIP, the FairBanks Community Health Intelligence Platform: community screening, referral and follow-up data capture, database, cloud sync and reporting dashboards, delivered across web and mobile.
- Building a cross-sector data collection and reporting application that writes the report as the user works, instead of leaving it to be compiled afterwards. Aimed at accounting, research, biology, agriculture, social media and similar fields.
- Developed the oxygen consumption formulas used to calculate patient demand during the COVID-19 response at International Hospital Kampala, and in support of Ministry of Health oxygen planning. The same method fed into capacity planning for later facility projects.
- Designed, built and deployed an AI reception agent for a law firm in Texas, USA. It handles live inbound client calls, books and reschedules appointments, and runs client follow-up. In active production use.
- Delivered hospital technology projects in five countries: Uganda, the Democratic Republic of Congo, Kenya, Tanzania and Somaliland. This included laboratory, theatre and radiology installations at the hospital serving the Kibali Gold Mine, DR Congo.
- Prepared and maintained the biomedical compliance evidence that carried International Hospital Kampala through successful COHSASA accreditation, and cut equipment downtime with a documented preventive maintenance programme.
- Upgraded partner health facilities across East and Central Africa for the Gould Family Foundation, including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi.

TECHNICAL SKILLS

Programming languages	Java · Python · JavaScript · TypeScript · C · C++ · C# · PHP · SQL · MATLAB · Arduino C · HTML5 · CSS3
Web and mobile	React · Next.js (App Router) · React Native · Expo · Tailwind CSS · styled-components · Bootstrap · JavaFX / FXML · Jest · Playwright
Backend and APIs	Node.js · Express · Laravel · CodeIgniter · Spring / Spring Boot · Convex · Prisma · REST APIs · JWT · role-based access control
Databases and data	MySQL · relational modelling and query optimisation · SQL analytics · data cleaning and reporting · MATLAB modelling · Excel analysis and dashboards
Cloud and DevOps	AWS · Microsoft Azure · Vercel · EAS · Bunny.net CDN · Git and GitHub · CI/CD pipelines · release and environment management
AI, agents and automation	AI agent design, deployment and monitoring · workflow and process automation · third-party systems integration · automated auditing and verification tooling · Cursor · Claude · OpenAI GPT · Codex
Operating systems	Windows · Linux · Android. Administration, deployment and troubleshooting on all three.
Engineering and scientific	Embedded systems design · biosignal processing · medical imaging · biomedical modelling and simulation · bioinformatics and functional genomics · Proteus · Eagle CAD · Solid Edge

PROFESSIONAL EXPERIENCE

Lead Software Developer, FCHIP and Health Information Systems

2026 to present

FairBanks Medical Centre Limited, Kampala, Uganda

- Lead the design and build of the FairBanks Community Health Intelligence Platform (FCHIP): data capture form design, database structure, cloud data flows, reporting dashboards, and delivery across both web and mobile.
- Building a cross-sector data collection and reporting application that turns work already being done into finished reports in real time, for accounting, research and laboratory science, biology, agriculture, social media and similar record-heavy fields.
- Architect and maintain the hospital and clinic software used daily by clinical and administrative staff, covering patient records, scheduling, referral tracking and management reporting.
- Design and deploy AI agents and automation pipelines that remove manual handling from reception, appointment management, client follow-up and routine reporting.
- Own technical documentation, data-protection practice, access control and release management across the platform.
- Write and edit institutional proposals, grant applications, partnership documents and investment materials. Chaired the FairBanks meeting with the Uganda Health Partners Cooperative on cooperative health insurance in July 2026.

Consultant Biomedical Engineer, Independent Practice

Mar 2025 to present

Contract assignments: Uganda, DR Congo, Kenya, Tanzania, Somaliland

- Install, commission, calibrate and maintain hospital medical equipment under contract across five countries, working independently on site and to a fixed handover standard.
- Completed laboratory, theatre and radiology installations at several hospitals in the Democratic Republic of Congo, including the facility serving the Kibali Gold Mine.
- Train clinical users at handover and produce a written service report and maintenance record for every assignment, so the client keeps a verifiable equipment history.
- Handle preventive and corrective maintenance, on-site fault diagnosis and specification advice for procurement.

Biomedical Programs Manager

Aug 2024 to Feb 2025

Gould Family Foundation (GFF), East and Central Africa

- Managed the regional biomedical team, covering programme planning, implementation and reporting across partner health facilities in multiple countries.
- Oversaw procurement, installation and commissioning of medical technologies, with programme documentation kept in line with healthcare technology standards.
- Developed equipment lifecycle-management and master-planning strategies that were applied across partner facilities.
- Upgraded several facilities, including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi, where oxygen and equipment demand modelling informed the capacity plan.
- Led biomedical and clinical user training programmes for engineers, technicians and clinical staff.

Biomedical Manager

Jan 2020 to Jan 2024

International Hospital Kampala (IHK)

- Managed the biomedical engineering function of a major private hospital for four years, including team supervision, workplans and reporting to executive management.
- Led oxygen therapy management through the COVID-19 response and supported Ministry of Health oxygen planning. Developed the formulas used to calculate patient oxygen consumption, which gave managers a dependable basis for sizing plant capacity, cylinder stock and flow requirements rather than estimating them.
- Led installation and commissioning of CT and X-ray imaging, oxygen plant, laboratory, theatre and ICU systems, through to handover and trained clinical users.
- Built the preventive maintenance schedules and service-record systems that reduced equipment downtime and kept devices safe for clinical use.
- Coordinated procurement, technical specification, commissioning documentation and lifecycle management of hospital medical equipment.
- Prepared and maintained the compliance evidence files reviewed for COHSASA accreditation, supporting the hospital through successful accreditation.

Biomedical Engineer

Apr 2019 to Jan 2020

Norvik Hospital Ltd, Kampala

- Maintained and serviced hospital diagnostic and patient-monitoring equipment.
- Supported installation and calibration of medical imaging and ICU equipment.
- Helped establish preventive maintenance schedules and service follow-up records.

Research Intern

Dec 2018 to Apr 2019

Uganda Virus Research Institute (UVRI)

- Contributed to biomedical research projects and laboratory experimentation.
- Carried out laboratory data analysis, mathematical modelling and research pipeline design in Python.

Teaching Assistant

Mar 2016 to Aug 2017

Makerere University, College of Health Sciences

- Assisted in teaching biomedical engineering course units at undergraduate level.
- Supervised and supported laboratory practical sessions for engineering students.

SELECTED PROJECTS

FairBanks Community Health Intelligence Platform (FCHIP)

2026 to present

Lead Software Developer · FairBanks Medical Centre Limited

Community, screening, referral and follow-up data capture; database structure; cloud data flows; management dashboards; web and mobile delivery. Built with Next.js, TypeScript, React Native and cloud data services.

Cross-sector data collection and reporting application

2026 to present

Founder and Lead Developer · In development

An application that simplifies data collection and reporting for professions that live on records: accounting, research, biology and laboratory science, agriculture and social media among them. Rather than collecting data now and writing the report later, the user works inside the app and the information and reports are generated in real time as they go.

COVID-19 oxygen therapy management and demand modelling

2020 to 2022

Biomedical Manager · International Hospital Kampala, with Ministry of Health planning support

Ran oxygen therapy management for the hospital through the pandemic and developed the formulas for calculating patient oxygen consumption. The figures were used to size plant capacity, cylinder stock and flow requirements, and to support planning decisions at managerial and national level.

AI reception agent

2025 to present

Designer and Developer · Law firm, Texas, USA

Built and deployed an AI agent that answers live inbound client calls, books and reschedules appointments, and runs post-contact client follow-up. In active production use.

Shulekeeper school information system

2021 to 2024

Assistant Software Developer · Shulekeeper

Multi-user records and reporting platform built with React and Next.js, TypeScript, Laravel, MySQL and AWS. Contributed front-end and back-end work, third-party integrations, testing, user training and documentation.

Endoscopy image capture and reporting software

2017 to 2019

Lead Software Developer · St. Catherine Hospital and St. Francis Hospital

Clinical image capture and procedure reporting system in JavaFX and MySQL. Led algorithm design, relational modelling, legacy data migration, on-site installation, user training and the user manual.

Wekebere foetal heart-rate monitoring system

2017 to 2024

Electronics Developer and Programmer · Wekebere

Android and Arduino monitoring system for third-trimester foetal heart-rate measurement, with Java serial communication between device and phone and Microsoft Azure cloud storage.

PROFESSIONAL COMPETENCIES

Programme and project management	Managed a regional biomedical team across East and Central Africa and a hospital biomedical function for four years: workplans, schedules, procurement coordination and management reporting.
Monitoring, evaluation and data	Leads the FCHIP indicator, data-quality, dashboard and reporting build. Built equipment registers, maintenance schedules and routine management reports. Analysis in Python, SQL, Java, JavaScript and MATLAB.
Proposal, grant and technical writing	Writes and edits institutional proposals, grant applications, investment materials, partnership documents, technical documentation and user manuals.
Data analysis and automated auditing	Cleans, models and reports on operational and clinical data. Builds automated auditing and verification tooling to check records in bulk rather than by hand.
Procurement, specification and compliance	Coordinated specification, procurement, commissioning and lifecycle documentation, and kept the compliance evidence reviewed for COHSASA accreditation.

Stakeholder and partnership management	Chairs partner meetings and writes them up. Works with hospital departments, suppliers, trainees, accreditation assessors, donors and prospective partners.
Training and capacity building	Trained biomedical engineers, technicians and clinical users across multiple facilities, and taught undergraduate engineering laboratory practicals at Makerere University.
Communication and adaptability	Clear written and spoken English, comfortable presenting to clinical, executive and donor audiences. Settles quickly into new countries, teams, codebases and equipment brands.

MEDICAL EQUIPMENT EXPERIENCE

Radiology and imaging	CT scanners · X-ray machines · ultrasound systems · C-arms
Theatre and intensive care	Theatre systems · ventilators · patient monitors · infusion pumps · critical-care devices
Oxygen and medical gas	Oxygen plants · concentrators · cylinder and manifold systems · piped medical gas · oxygen demand and capacity calculation
Laboratory	Haematology analysers · clinical chemistry analysers · blood bank and transfusion equipment
Maternity and neonatal	Neonatal incubators · phototherapy units · foetal Dopplers · CTG machines
Support plant	RO water treatment systems · sterilisation and autoclave equipment

EDUCATION

Certificate in Leadership and Management in Health

2022

University of Washington

Grade A+. Planning, leadership and management of health programmes.

BSc Biomedical Engineering, Second Upper Honours

2012 to 2017

Makerere University, Kampala

Key course units: Information and Communication Technology, Computer OOP Principles, Software Engineering, Database Systems, Computer-Based Medical Systems, Embedded Systems, Biosignal Processing and Analysis, Medical Imaging, Biomedical Modelling and Simulation, Bioinformatics and Functional Genomics, Clinical Engineering, Research Methods, Principles of Management.

Advanced Java Programming, Level 1

2015 to 2016

Green Bridge School of Open Technologies

Grade A+.

Uganda Advanced Certificate of Education (UACE)

2011

Mengo Secondary School

Principal subjects: Mathematics, Physics, Biology, Chemistry.

Uganda Certificate of Education (UCE)

2009

St. John's Wakiso Secondary School

TRAINING AND CERTIFICATION

- **COHSASA accreditation standards: in-service preparation and compliance.** International Hospital Kampala, 2020 to 2024.
- **Fire Safety, Prevention, Firefighting and Emergency Scene Management.** Fire Technologies Limited (IHK), 2021.

INTERESTS

Music, and playing the saxophone in particular. · Film. · Physical fitness and regular exercise. · Travel and working in unfamiliar places, which the regional field work has made second nature.

REFEREES

Eng. Richard Ssejongo

Biomedical Engineer, St. Francis
Hospital Nsambya

+256 753 818 754 · +256 777 132 489

Dr. Annet Khingi

Senior Radiologist and Administrator,
Mengo Hospital

+256 772 592 771 · +256 701 592 771

Racheal Nabukeera

Director and Founder, FairBanks
Medical Centre

+256 772 849 258 ·
nracheal017@gmail.com